

Media Bias Detection in Political News

A Machine Learning Approach | CSC680: Data Mining | Spring 2026

Alyssa Langmeyer

The Question

Can a machine learn to spot biased news just from how it's written?

Not by knowing the outlet or author. Just from the words on the page.

The Dataset

NewsMediaBias-Plus | Vector Institute, 2024

40,945

news articles

91

news outlets

69% / 31%

biased vs. unbiased

Challenge: the 69/31 imbalance means a lazy model could hit 69% accuracy just by always predicting biased so I had to build in safeguards.

How It Works

1

Clean the Text

Scraped articles come with a lot of junk: cookie banners, subscription prompts, timestamps. Strip all of that — it's noise, not content

Combine the headline and body into one text field (headlines show slant more clearly than the body)

Remove the outlet's name from the text. If 'Reuters' stays in, the model just learns 'Reuters = unbiased' without understanding why. Removing names forces it to judge writing style alone

Drop anything under 50 characters — those are failed scrapes with no content

2

Convert to Numbers

ML models can't read words — everything has to become a number

TF-IDF scores each word by how distinctive it is to this article. 'The' appears everywhere so it scores near zero. 'Assassination attempt' in one article scores high

Also looks at two-word pairs: 'far right' tells you far more than 'far' or 'right' alone

Each article becomes a list of 10,000 scores, one per word or phrase

3

Train and Test

70% of articles used to learn patterns, 30% held back to test on — model never saw the test set during training

The dataset is 69% biased, so without a correction the model could just always guess 'biased' and be right 69% of the time without learning anything useful

Added a penalty so both classes get equal weight during training

Chose Logistic Regression because you can see exactly which words pushed each prediction

4

Dig Deeper

Ran extra tests to understand what the model actually learned, not just whether it was accurate

Which individual words are most statistically different between biased and unbiased articles?

What's the average bias score for each of the 91 outlets?

Can articles cluster into biased/unbiased groups with no labels at all? (No — they group by topic instead, which is itself a useful finding)

Media bias is about writing style, not which political side you're on.

Bias is detectable from text alone, without knowing who published it. The model never sees the outlet name, author, or publication date, just the words. That's the whole point: if it works, it's learning what biased writing actually looks like.

Opinionated writing has a recognizable style regardless of political direction. Left-leaning and right-leaning advocacy outlets both use charged adjectives, evaluative labels, and framing language. Wire services like AP don't and that stylistic difference is what the model picks up on.

You can't find bias by clustering articles without labels. When you let the algorithm group articles on its own, it organizes them by topic — Olympics articles together, Supreme Court articles together — not by writing style. Labeled training data is necessary. Bias is a subtler signal than subject matter.

Removing outlet names from the text was the single most consequential decision in the whole project. If "Reuters" stays in the article, the model learns "Reuters = unbiased" and stops there. Stripping names forced it to learn from how the article is written, not who wrote it.

How Well Did It Work?

ROC-AUC

0.82

Correctly ranks a biased article above an unbiased one 82% of the time. Random guessing = 0.50.

F1 (biased class)

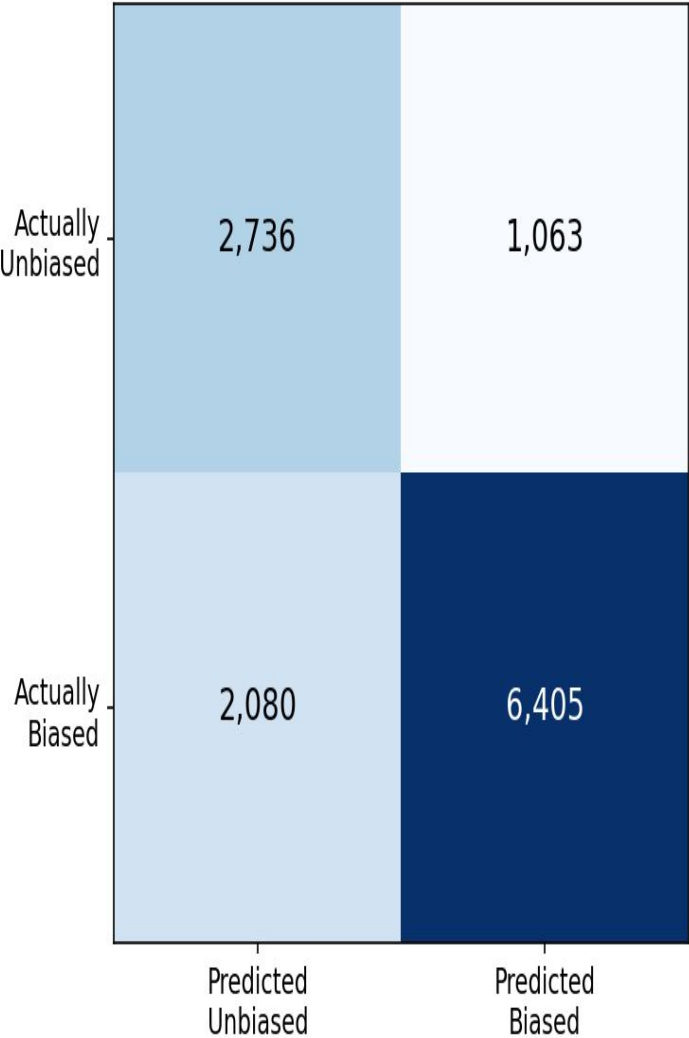
0.80

F1 (unbiased class)

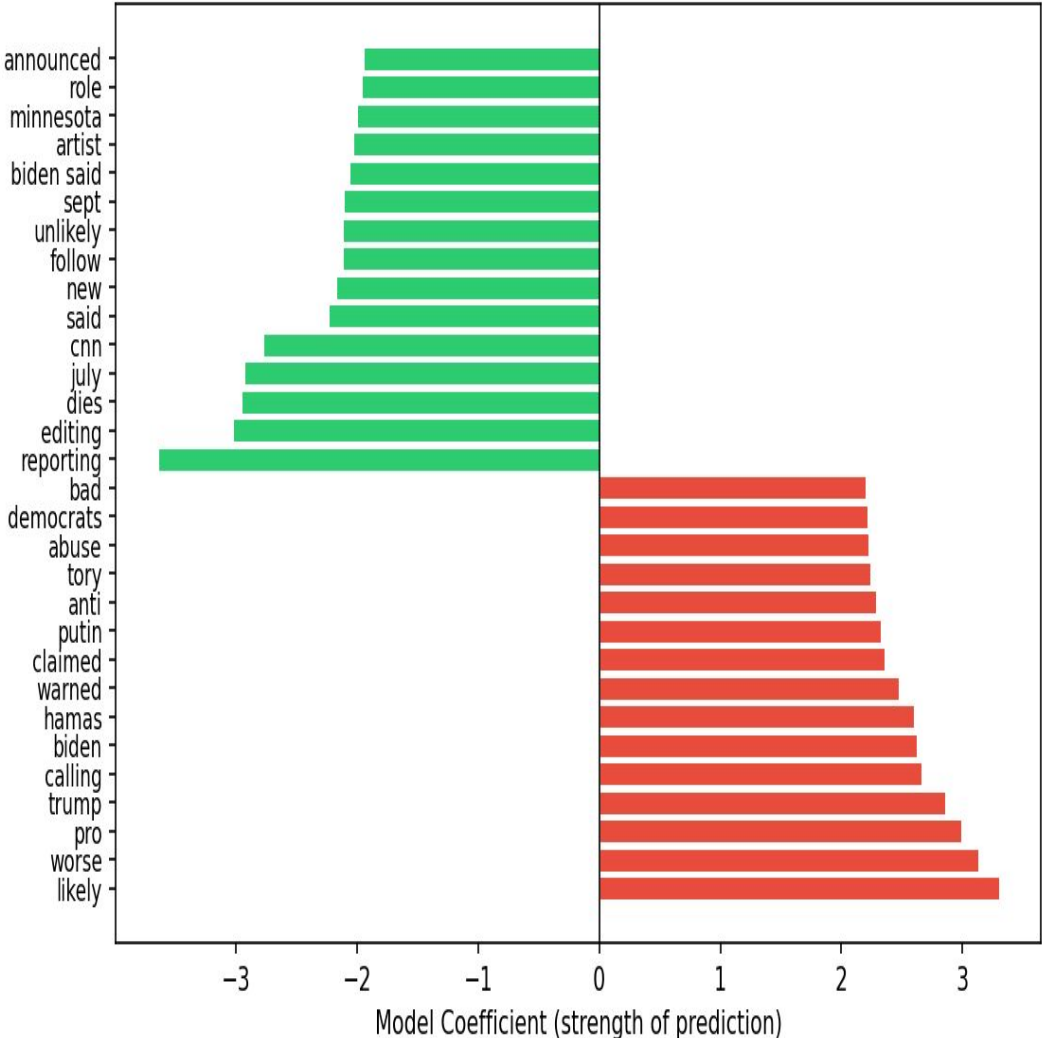
0.64

Lower unbiased F1 reflects the 69/31 imbalance, not a broken model.

Confusion Matrix



Top Words: Biased (red) vs. Unbiased (green)



What Words Give It Away?

Predicts BIASED

Trump

Biden

Harris

far right

White House

assassination attempt

Supreme Court

radical

Predicts UNBIASED

"said"

"police said"

"year old"

"gold medal"

"paris olympics"

"high school"

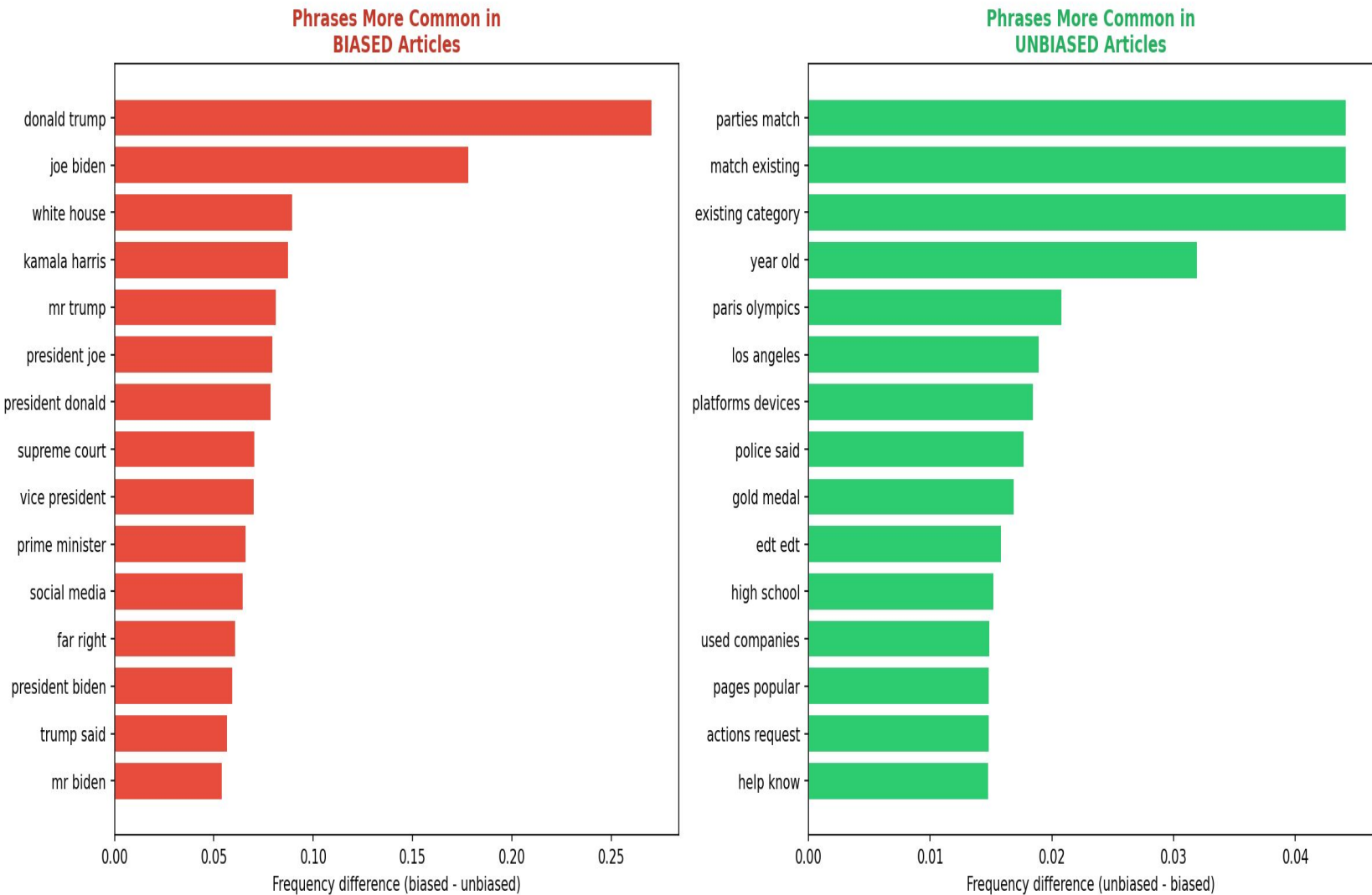
"announced"

"according to"

Bias shows up in word choice, not just topic. Two outlets can cover the same event with completely different language.

Who Scores Most Biased?

Bigram Analysis: Distinguishing Phrases by Label



The surprising part

Most biased outlets span both sides of the spectrum. The Federalist and Daily Caller (right) sit next to ProPublica, Daily Kos, and The Guardian US (left).

What they share

Opinionated, advocacy-driven writing style. Not political affiliation.

Least biased

AP, Reuters, CNBC, Bloomberg. Wire services and financial news. Neutral and fact-forward.

The real finding

This dataset measures opinionated vs. neutral writing, not left vs. right politics.

Where to Go From Here

Fine-tune BERT

Train DistilBERT end-to-end on bias classification instead of frozen embeddings. Expected improvement of 5+ points on AUC.

Topic-stratified analysis

Does bias look different in immigration stories vs. economic stories? Some topics may be more prone to opinionated framing.

Neutral rewriting

Use bias predictions to guide a generative model to rewrite flagged articles in neutral wire-service style.

Multi-class bias

The dataset also labels political, gender, and racial bias separately. A multi-class model would give more specific outputs.